Relapse Thresholds in Quiescent Cancer Cells Dormancy

A Delay Differential Equation Model for Tumor Growth

by

Jaxen Anirban Dutta
jaxen.dutta@uwaterloo.ca
Sabeer Ghura
sghura@uwaterloo.ca
Inchang Huh
ihuh22@uwaterloo.ca

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Table of Contents

Acknowledgements **ii**

List of Figures **v**

List of Tables **vi**

List of Abbreviations **vii**

1 Introduction **1**

1.1 Background 1

1.2 Why a DDE Model for Tumor Growth? 2

2 The Existing Delay Differential Equation Model **4**

2.1 Defining the Model 4

2.1.1 Assumptions 4

2.1.2 The DDE Model 5

2.1.3 Model Parameters 5

2.2 Strengths and Weaknesses of the Model 7

2.2.1 Strengths: Bridging the Gap Between Biological Reality and Mathematical Modelling 7

2.2.2 Weaknesses: Addressing Complexity and Clinical Applications . . . 8

2.2.3 Future Directions: Building Upon a Strong Foundation 9

3 Implementation of Existing Model **10**
3.1 Software Selection 10
3.2 Model Implementation 10
3.3 Model Simulation Visualization 10

4 Analysis of Existing Model **12**
4.1 Published Stability Analysis 12
4.2 Extension: Numerical Stability Analysis 13

5 Extension: Incorporation of QCCs **14**
5.1 Extended Model Definition 14
5.1.1 Species and Parameter Extensions 15
5.1.2 DDE Model Update 15
5.2 Analysis 17
5.2.1 Behaviour Compared to Original Model 17
5.2.2 Effect of aq and dq 17
5.2.3 Effect of chemotherapy schedule 19

6 Conclusion **21**

References **22**

APPENDICES **24**

A Python Code **25**

Glossary **28**

List of Figures

3.1 Typical simulations showing **(a)** exponential growth and **(b)** containment by the immune system 11

4.1 Basin of Attraction 13

5.1 Simulation of a Chemotherapy Infusion **(a)** without and **(b)** with the Quiescent Cancer Cells (QCC) sub-population 17

5.2 Simulation of a chemotherapy with different values of a_q and d_q can result in the tumor population **(a)** being eradicated or **(b)** surviving 18

5.3 Basin of Attraction for extended model 18

5.4 Simulation of chemotherapy with 3 days of medication followed by 3 days of rest 19

5.5 Simulation of chemotherapy with 3 days of medication followed by 4 days of rest 20

List of Tables

2.1 Parameter values used in [1] (non-dimensional) 6
2.2 Description of key model parameters 6

List of Abbreviations

G_0 Quiescent Phase 1, 5, 15

G_1 Growth Phase 1, 2

G_2 Preparation for Mitosis Phase 1

M Mitosis Phase 1

S DNA Synthesis Phase 1

DDE Delay Differential Equation 1–3, 5, 7, 9, 10, 21

ODE Ordinary Differential Equation 2, 3

QCC Quiescent Cancer Cells 14–17, 21

Chapter 1

Introduction

Cancer's relentless growth poses a constant challenge to human health. In Canada alone, projections estimate nearly 90,000 people aged 35 and above will lose their lives to this disease in 2024, with numbers expected to rise significantly [2]. This grim reality underscores the critical need for advancements in cancer research, particularly in understanding the growth dynamics of solid tumors.

Our report addresses this need by delving into the world of mathematical modelling. We will focus on a Delay Differential Equation (DDE) model to examine how tumor cell populations interact with cytotoxic drugs. By analyzing this interaction, we hope to gain valuable insights into developing effective treatment strategies and identifying thresholds for cancer cell relapse.

1.1 Background

This section provides a foundational understanding of three key areas relevant to our study.

1. ***The Cell Cycle:*** Cells undergo a tightly regulated cycle of growth, replication, and division. The cell cycle comprises four main phases: Growth Phase (G_1), DNA Synthesis Phase (S), Preparation for Mitosis Phase (G_2), and Mitosis Phase (M). The Quiescent Phase (G_0) represents a temporary resting state cells can enter and exit. Disruptions in cell cycle regulation can lead to uncontrolled cell proliferation, a hallmark of cancer [3].

2. ***Tumor Growth:*** Tumors arise from abnormal cell growth and division. Understanding the dynamics of tumor growth, including cell proliferation, death, and interaction with the immune system, is essential for modelling treatment effectiveness. Mathematical models offer valuable insights into these complex dynamics [4].

3. ***DDE:*** DDEs are a specific class of differential equations that incorporate time delays into the model. These delays represent the time it takes for a process to occur, such as the time it takes a cell to transition from G_1 to mitosis [5]. DDEs are particularly well-suited for modelling biological systems with inherent time delays, such as tumor growth where cell division takes time.

By integrating these concepts, we establish the foundation for developing a delay differential equation model that captures the dynamics of tumor growth and treatment response. This model can be used to analyze treatment strategies and predict their impact on tumor progression.

1.2 Why a DDE Model for Tumor Growth?

Conventional mathematical models of tumor growth often use Ordinary Differential Equation (ODE)s that capture the rates of change in cell populations. However, ODEs have limitations when representing biological systems with inherent time delays. In tumor growth, these delays arise from various biological processes, making DDEs a more suitable modelling tool. The following are some reasons why DDEs are advantageous for this specific application:

1. ***Incorporating Cell Cycle Dynamics:*** The cell cycle, a tightly regulated process of growth and division, plays a crucial role in tumor development. DDEs can explicitly account for the time it takes for cells to transition between different phases within the cell cycle, allowing for a more realistic representation of tumor growth compared to ODE models.

2. ***Modeling Drug Effects:*** Certain anti-cancer drugs target specific stages of the cell cycle. DDEs can capture the time delay between drug administration and its impact on cell proliferation. This allows for a more accurate assessment of treatment efficacy and the potential emergence of drug resistance due to time-dependent effects [6].

3. ***Understanding Immune System Response:*** The immune system plays a critical role in controlling tumor growth. DDEs can be used to model the time delays associated with immune cell activation, recognition, and attack on tumor cells. This allows for a more comprehensive analysis of the interplay between tumor dynamics and the immune response [7].

By incorporating time delays, DDE models provide a more nuanced and realistic representation of tumor growth compared to simpler ODE models. These models have the potential to be valuable tools for researchers and clinicians studying tumor biology, treatment strategies, and personalized medicine approaches.

Chapter 2

The Existing Delay Differential Equation Model

This chapter details a mathematical model for tumor growth originally proposed by Villasana and Radunskaya (2003) [8]. The model incorporates delay differential equations to represent the time it takes for tumor cells to transition through different phases of the cell cycle.

2.1 Defining the Model

The model tracks four key populations:

- $T_I(t)$: Tumor cells in interphase (actively growing but not undergoing mitosis).
- $T_M(t)$: Tumor cells in mitosis (actively dividing).
- $I(t)$: Immune cells responsible for attacking tumor cells.
- $u(t)$: Drug concentration acting on the tumor cells and immune cells.

2.1.1 Assumptions

All mathematical models of biological phenomena are simplifications of real-world systems. Here, we outline the key assumptions underlying the model described in [8]:

- ***Instantaneous Mitosis:*** Mitosis, the cell division process, is assumed to be instantaneous compared to the overall cell cycle duration.
- **Drug Targeting Mitosis:** The drug is modelled to specifically target and arrest tumor cells during mitosis. This arrest ultimately leads to cell death. Note that the authors use growth arrest and cytotoxicity interchangeably in this context.
- **Drug Impact on Both Cell Types:** The drug is assumed to affect both tumor cells and immune cells. This is because some chemotherapy drugs can be cytotoxic to rapidly dividing cells, including those in the bone marrow where immune cells are produced. [8]
- **Exclusion of G_0 Phase:** The model does not explicitly consider a subpopulation of tumor cells residing in a dormant state, known as the G_0 phase. These quiescent cells are less susceptible to most chemotherapeutic agents and are estimated to comprise roughly 20% of the total tumor cell population [8]. Excluding this phase is a simplification, and the authors acknowledge that its inclusion could be valuable for future exploration.
- **Exponential Drug Decay:** Drug concentration is assumed to decay exponentially over time, a common characteristic of many drugs with a single half-life parameter.

2.1.2 The DDE Model

The model is represented by a system of DDEs:

$$
\begin{aligned}
T_I' &= 2a_4T_M - (c_1I + d_2)T_I - a_1T_I(t-\tau) \\
T_M' &= a_1T_I(t-\tau) - d_3T_M - a_4T_M - c_3T_MI - k_1(1-e^{-k_2u})T_M \\
I' &= k + \frac{\rho I(T_I+T_M)^n}{\alpha + (T_I+T_M)^n} - c_2IT_I - c_4T_MI - d_1I - k_3(1-e^{-k_2u})I \\
u' &= -\gamma u
\end{aligned}
$$

2.1.3 Model Parameters

Table 2.1 shows the parameter values used in [1]. It is important to note that these values are non-dimensional.

$\tau = 0.92$	$a_1 = 0.98$	$a_4 = 0.8$	$d_2 = 0.11$
$d_3 = 0.4$	$d_1 = 0.29$	$c_1 = 9 \cdot 10^{-1}$	$c_3 = 9 \cdot 10^{-1}$
$c_2 = 85 \cdot 10^{-3}$	$c_4 = 85 \cdot 10^{-3}$	$\rho = 0.1$	$\alpha = 0.2$
$k = 0.036$	$n = 3$	$\lambda_1 = 126.12$	$\lambda_2 = 0.85$
$k_1 = 0.47$	$k_2 = 0.57$	$k_3 = 0.49$	$k_4 = 0.061$
$r_1 = 0.73$	$r_2 = 0.27$		

Table 2.1: Parameter values used in [1] (non-dimensional)

We are using the parameter values from Villasana and Ochoa's 2004 paper [1], as Villasana and Radunskaya's 2003 paper did not include parameters related to drug interaction. [8]

The table below provides a brief description of some key parameters used in the model:

#	Symbol	Description
1	τ	Time spent by cells in interphase.
2	a_i	Rates of cell reproduction.
3	d_i	Rates of natural cell death.
4	c_i	Rates of loss due to tumor-immune cell interaction.
5	ρ, α, n	Parameters for the Michaelis-Menten term describing immune cell growth stimulated by tumor cells.
6	k	Basal growth rate of immune cells.
7	k_i	Parameters related to the drug's effect on tumor and immune cells.
8	γ	Drug decay rate.
9	λ_i	Parameters representing the decay rates of the two drug terms.
10	r_i	Parameters representing the proportions of the two drug terms.

Table 2.2: Description of key model parameters

Note: The original model in [8] used a single drug term instead of λ_i and r_i. This new set of parameters was used in [1] where the authors use a modified model with two drug terms. For our model [8], we elect to use $\gamma := \lambda_2$.

By analyzing these equations and parameter definitions, we can gain a deeper understanding of the biological processes. The following is a deconstruction of the model equations (2.1.2) to further clarify their biological meaning.

$$T_I' = \underbrace{2a_4T_M}_{\substack{\text{Cells entering}\\\text{Interphase}\\\text{after Mitosis}}} - (\underbrace{c_1I}_{\substack{\text{Combat}\\\text{Deaths}}} + \underbrace{d_2)\,T_I}_{\substack{\text{Natural}\\\text{Deaths}}} - \underbrace{a_1T_I(t-\tau)}_{\substack{\text{Interphase Cells}\\\text{entering Mitosis}}}$$

$$T_M' = \underbrace{a_1T_I(t-\tau)}_{\substack{\text{Interphase Cells}\\\text{entering Mitosis}}} - \underbrace{d_3T_M}_{\substack{\text{Natural}\\\text{Deaths}}} - \underbrace{a_4T_M}_{\substack{\text{Cells entering}\\\text{Interphase}\\\text{after mitosis}}} - \underbrace{c_3T_MI}_{\substack{\text{Combat}\\\text{Deaths}}} - \underbrace{k_1(1-e^{-k_2u})T_M}_{\text{Drug Interaction}}$$

$$I' = \underbrace{k}_{\substack{\text{Basal}\\\text{Growth Rate}}} + \underbrace{\frac{\rho I(T_I+T_M)^n}{\alpha+(T_I+T_M)^n}}_{\substack{\text{Growth due to stimulus}\\\text{(Michaelis-Menten)}}} - \underbrace{c_2IT_I}_{\substack{\text{Combat}\\\text{Deaths}}} - \underbrace{c_4T_MI}_{\substack{\text{Combat}\\\text{Deaths}}} - \underbrace{d_1I}_{\substack{\text{Natural}\\\text{Deaths}}} - \underbrace{k_3(1-e^{-k_2u})I}_{\text{Drug Interaction}}$$

$$u' = -\underbrace{\gamma u}_{\text{Exponential Decay}}$$

2.2 Strengths and Weaknesses of the Model

This paper offers a novel application of DDEs to model tumor growth, incorporating the crucial factor of cell cycle delays. While this approach provides a fresh perspective on tumor dynamics (unlike simpler models that neglect time lags), it also presents limitations that warrant further exploration. Here, we delve into the model's strengths and weaknesses.

2.2.1 Strengths: Bridging the Gap Between Biological Reality and Mathematical Modelling

A key strength lies in its biological relevance. By explicitly considering cell cycle delays, the DDE approach offers a more realistic portrayal of tumor growth compared to simpler models. This enhanced relevance allows for a more accurate representation of real-world scenarios.

Furthermore, the paper establishes a robust analytical framework. The usage of various equations, and theorems, and numerical tools provides a mathematical link between model

parameters and tumor-free fixed point stability. This empowers researchers to investigate how factors like treatment efficacy or immune response can influence tumor behavior. The ability to predict stability based on parameters allows for targeted control strategies. For instance, researchers can identify optimal drug concentrations that promote tumor regression.

Beyond stability, the paper explores periodic solutions using the argument principle and Hopf bifurcation analysis. This opens doors to investigating potential causes of fluctuating tumor sizes observed clinically. By understanding conditions leading to such patterns, researchers can gain insights into tumor heterogeneity and develop tailored treatment strategies.

2.2.2 Weaknesses: Addressing Complexity and Clinical Applications

The model's generalizability to different tumor types might be limited. Tumors exhibit diverse growth patterns depending on the specific cancer and involved mutations. The current model might require refinement to capture the dynamics of different sub-types.

The use of one delay variable limits the inclusion of more delays for cell phases and cell-cycle-specific drug interaction. Additionally, the model simplifies the immune system and ignores drug resistance within the tumor cell population. It also assumes a deterministic system, whereas some components may be more stochastic.

Another limitation is the lack of data validation. The strong theoretical framework requires validation with real-world tumor growth data for broader acceptance and practical application. Fitting model parameters to patient data and comparing predicted growth patterns with observed data would strengthen the model's credibility and enhance its value in clinical settings.

Finally, the limited scope of the analysis presents an opportunity for further exploration. While the focus on fixed point stability and periodic solutions provides valuable insights, a more comprehensive understanding can be achieved by exploring the model's behavior under a broader range of parameter values or incorporating additional biological factors like nutrient availability, spatial heterogeneity, or the immune system's influence.

2.2.3 Future Directions: Building Upon a Strong Foundation

By addressing the limitations outlined above, the model's robustness and applicability can be significantly enhanced:

1. ***Model Refinement:*** Refining the model to account for different cell cycle lengths or incorporating additional delays could increase its generalizability. This could involve incorporating parameters reflecting variations in cell cycle duration observed in different cancers.

2. ***Data Integration:*** Integrating patient data for parameter estimation and model validation would strengthen the model's clinical relevance and allow for personalized treatment approaches. This would involve collaborating with clinicians to collect patient data and utilize statistical techniques to fit the model parameters to specific tumor characteristics.

3. ***Model Expansion:*** Extending the analysis to explore a broader range of parameter values and incorporating additional biological factors could lead to a more comprehensive understanding of tumor growth dynamics. This would involve developing computational tools to explore the model's behavior under various parameter combinations and incorporating additional biological modules that represent factors like nutrient availability and immune response.

This DDE model presents a valuable contribution by capturing cell cycle delays. The established analytical framework lays a strong foundation for further research. By addressing the limitations, researchers can significantly enhance this approach, ultimately contributing to personalized medicine and the development of more effective treatment strategies for cancer patients.

Chapter 3

Implementation of Existing Model

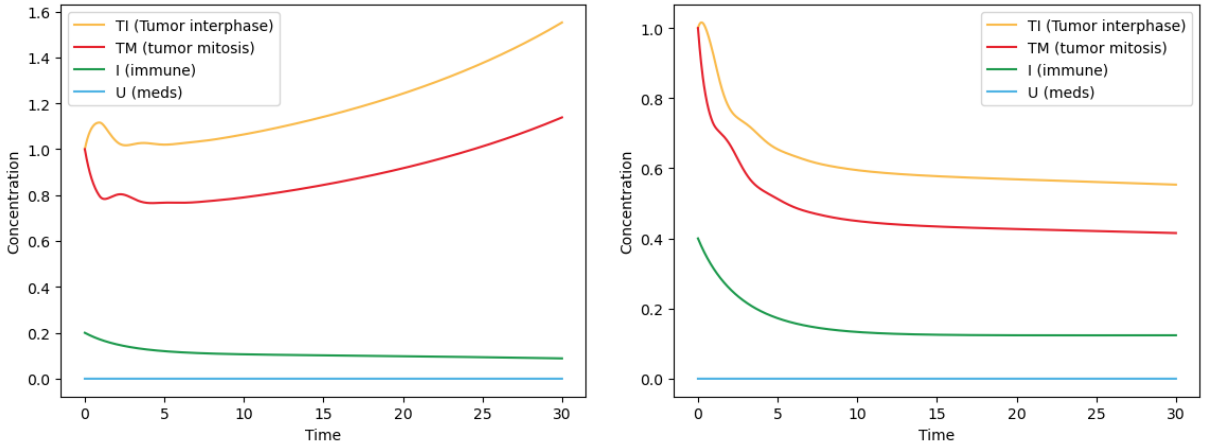
3.1 Software Selection

We chose to implement the model using `Python`. This decision was primarily driven by `Python`'s support of the `ddeint` library, which is a well-documented DDE solver. It also supports `pyplot`, which is a convenient way to plot simulation results. Sample code can be found in Appendix A.

3.2 Model Implementation

We chose to use a constant value for the history function of the DDE. Although this is not realistic, it is considered an acceptable simplification since $\tau = 0.92$ is considerably shorter than the simulation's duration [1].

3.3 Model Simulation Visualization



(a) Exponential growth

(b) Contained Growth

Figure 3.1: Typical simulations showing **(a)** exponential growth and **(b)** containment by the immune system

Chapter 4

Analysis of Existing Model

4.1 Published Stability Analysis

The paper performs stability analysis on the model described in 2.1.2. In general, there are multiple steady states for the system but the tumor-free steady state $(T_I, T_M, I, u) = (0, 0, k/d1, 0)$ is of particular interest since the is the goal of chemotherapy is to eradicate all cancer cells [8].

This tumour-free steady state always exists, but its stability depends on the exact values of the parameters. In particular, it can be shown that for certain values of parameters the stability of the tumor-free steady state depends on τ, where the steady state becomes unstable after a large enough threshold [8]. However, a finite number of stability switching can occur between $\tau = 0$ and the threshold [9]. This means that as τ increases, the system can switch between stable and unstable multiple times before it becomes permanently unstable. This is a non-intuitive result since one might expect that increasing τ would only make the system more unstable.

Many chemotherapy drugs work by trapping cancer cells in a particular phase in their cell cycle, where cells eventually die from natural causes. This can be interpreted as increasing τ. This analysis shows that caution must be taken when doing this since by changing the timing of the cell cycle, the tumor-free steady state might switch from stable to unstable. This would mean that when treatment is completed, the system would not move towards the tumor-free state [8].

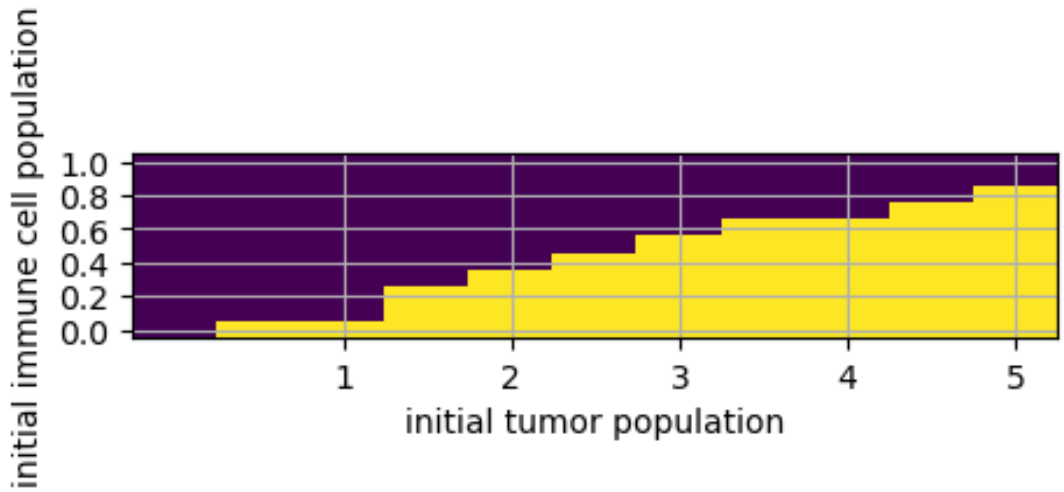


Figure 4.1: Basin of Attraction

4.2 Extension: Numerical Stability Analysis

A property of interest is the basin of attraction towards the tumour-free steady state. This is because once the system is within the basin of attraction, the system will converge to a tumor-free state even without drug intervention.

We can numerically estimate the basin of attraction by testing different initial conditions and plotting whether the system converges toward the tumour-free steady state. Figure 4.1 shows the estimated basin of attraction using the parameters in Table 2.1. The horizontal axis is the initial tumor population (split evenly between T_I and T_M) and the vertical axis is the initial immune cell population. The purple region is where the tumor population converges to the tumour-free steady state, and the yellow region is where it diverges. A simulation was considered to have converged if the total tumor population decreased by the end of the simulation.

A limitation of numerical estimation is that there may be false positives due to insufficient simulation duration or time step size.

Chapter 5

Extension: Incorporation of QCCs

Traditional mathematical models of tumor growth, like the one presented earlier, often focus solely on actively proliferating cancer cells. However, cancer cell populations are more heterogeneous, containing a subpopulation of QCCs in a dormant state. These QCCs are less susceptible to traditional therapies due to their low metabolic activity and can contribute to tumor recurrence [10]. To enhance the model's realism and capture this biological complexity, we propose extending the model by incorporating QCC dynamics.

5.1 Extended Model Definition

The existing tumor growth model is extended by incorporating the dynamics of QCCs. These QCCs are less susceptible to traditional therapies and can contribute to tumor recurrence. To achieve this, we modify the existing model equations to account for the following key processes:

1. ***Transition between Proliferating and Quiescent States:*** Two new parameter terms (a_q and d_q) are introduced representing the transition rates between the proliferating and quiescent states of cancer cells.

2. ***Quiescent Cell Proliferation:*** While QCCs are less active, they still retain the ability to proliferate under certain conditions. The model will account for this by including a proliferation term for the QCC population, T_Q.

3. ***Drug Sensitivity of QCCs:*** The model will consider the potential resistance of QCCs to a cytotoxic drug administered after a constant number of days.

5.1.1 Species and Parameter Extensions

The original model is expanded to account for the new QCC population and the associated parameters governing their dynamics.

Species Extension

T_Q, representing the population of tumor cells residing in the G_0 phase.

Parameter Extension

- a_q: Rate of production of QCCs from the proliferative cancer cell population (T_M).
- d_q: Rate at which QCCs transition back into the proliferative phase, effectively exiting the G_0 state.
- τ_q: Characteristic time delay associated with QCCs exiting the G_0 phase. This parameter accounts for the biological processes that need to occur before a QCC re-enters the cell cycle.

5.1.2 DDE Model Update

The original model's equations (2.1.2) are updated to incorporate the G_0 phase dynamics as follows:

$$T_Q' = \underbrace{2a_qT_M}_{\substack{\text{Cancer cells}\\ \text{entering } G_0}} - \underbrace{d_qT_Q(t-\tau_q)}_{\substack{\text{Cancer cells}\\ \text{exiting } G_0}} \tag{5.1}$$

$$T_I' = 2a_4T_M - (c_1I + d_2)T_I - a_1T_I(t-\tau) + \underbrace{d_qT_Q(t-\tau_q)}_{\substack{\text{Cancer cells}\\ \text{exiting } G_0 \text{ and}\\ \text{entering Interphase}}} \tag{5.2}$$

$$T_M' = a_1T_I(t-\tau) - d_3T_M - a_4T_M - c_3T_MI - k_1(1-e^{-k_2u})T_M - \underbrace{a_qT_M}_{\substack{\text{Cancer cells}\\ \text{entering } G_0}} \tag{5.3}$$

Equation 5.1 describes the change in the quiescent cell population (T'_Q). The first term ($a_q T_M$) represents the influx of cells from the proliferative population, while the second term ($d_q T_Q(t-\tau)$) accounts for cells exiting the quiescent phase with a time delay of τ.

Equations 5.2 and 5.3 are modified versions of the original model's equations for the interphase cell population (T_I) and the proliferative cancer cell population (T'_M), respectively. They now include a term ($d_q T_Q(t-\tau_q)$) representing the influx of QCCs that have exited the quiescent phase and entered interphase. Additionally, the term ($a_q T_M$) in the third equation captures the loss of proliferative cells due to their entry into the quiescent phase.

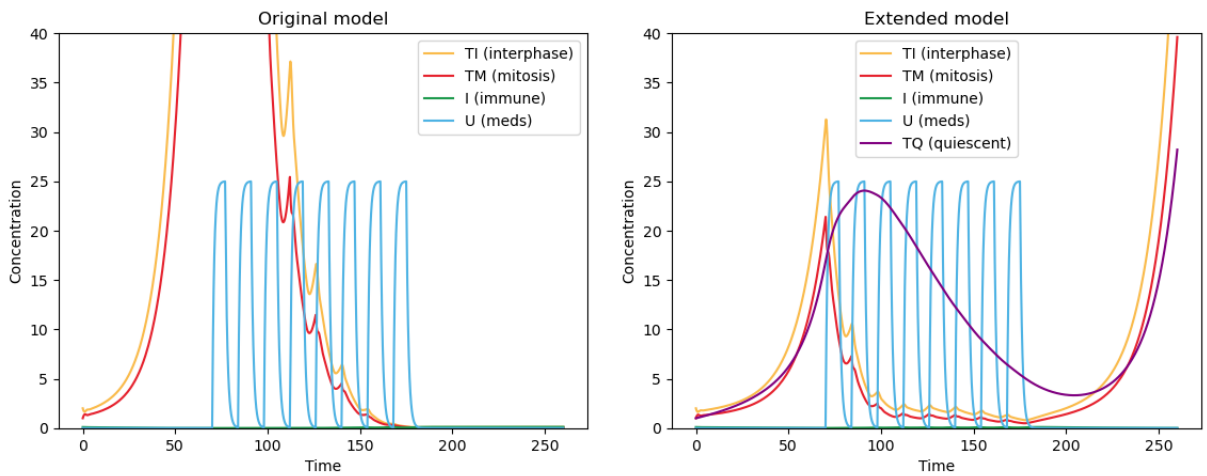
The differential equations for I and u remain identical to 2.1.2.

This extension allows the model to capture the dynamics of tumor cell populations that exhibit a quiescent phase, providing a more comprehensive representation of the underlying biological processes.

5.2 Analysis

5.2.1 Behaviour Compared to Original Model

As expected, the introduction of QCCs makes treatment more difficult, as shown in Figure 5.1 where an identical chemotherapy regimen fails to eradicate tumor cells when QCCs are introduced.



(a) Without QCCs

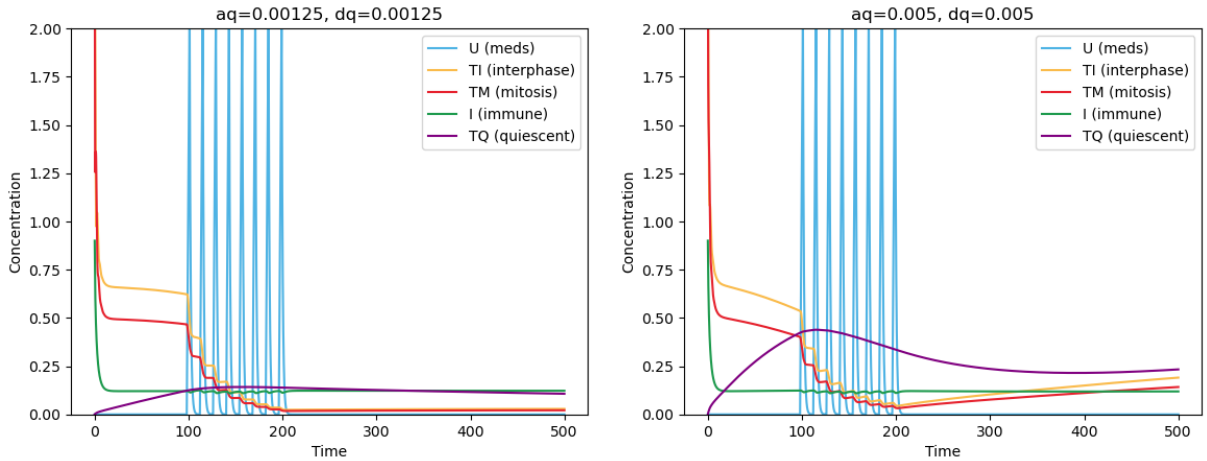
(b) With QCCs

Figure 5.1: Simulation of a Chemotherapy Infusion **(a)** without and **(b)** with the QCC sub-population

5.2.2 Effect of a_q and d_q

The cancer population's ability to evade chemotherapy depends on the values of a_q and d_q. An example of this phenomenon can be seen in Figure 5.2.

We can perform analysis similar to 4.2, where we test different combinations of a_q and d_q. Figure 5.3 shows the results of this analysis for our extended model. Again, the purple region is where the tumor population converges to the tumour-free steady state, and the yellow region is where it diverges. In this case, we considered a system to be inside the basin of attraction if the tumor population at the end of the simulation is smaller than immediately after the chemotherapy period.



(a) Tumor population decreases after chemotherapy (b) Tumor population increases after chemotherapy

Figure 5.2: Simulation of a chemotherapy with different values of a_q and d_q can result in the tumor population **(a)** being eradicated or **(b)** surviving

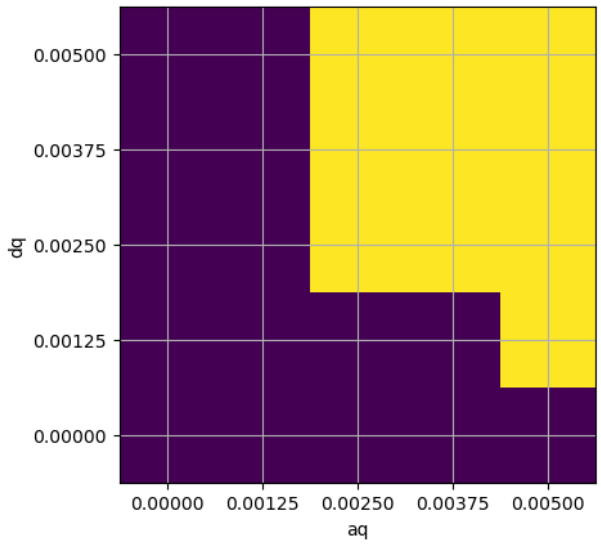


Figure 5.3: Basin of Attraction for extended model

We see from 5.3 that, for our particular choice of chemotherapy, values of a_q and d_q above 0.0025 result in the tumor population being able to survive chemotherapy.

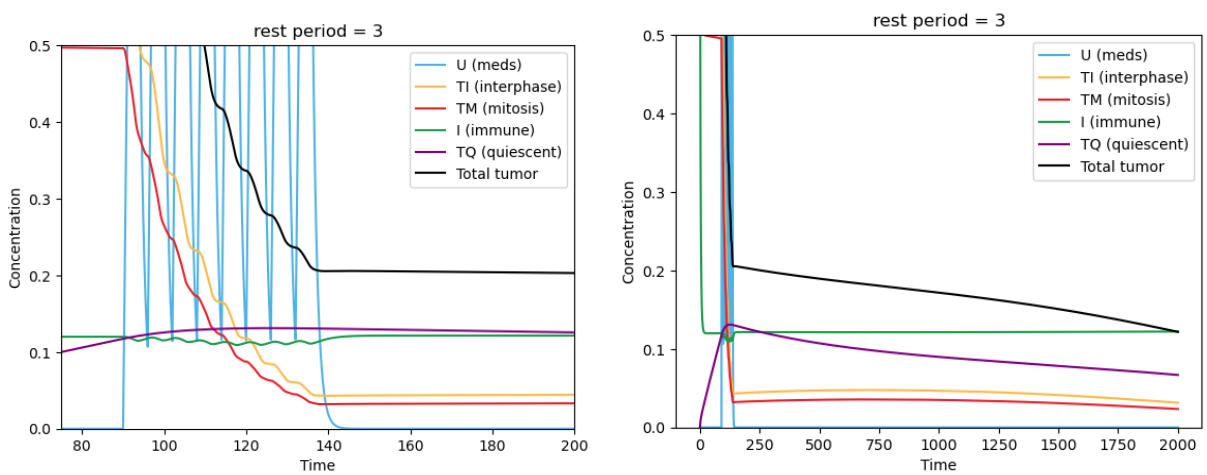
Simulation details for this analysis:

- $\tau_q = 30$ was arbitrarily chosen.
- The chemotherapy regiment simulated was $8\times$(3 days of chemotherapy +11 days of rest), where u reaches 0.7 (arbitrary choices).
- Initial conditions $(T_I, T_M, I, u, T_Q) = (2.5, 2.5, 0.9, 0, 0)$ was chosen arbitrarily.
- All other parameter values are identical to Table 2.1.

5.2.3 Effect of chemotherapy schedule

Another variable that influences whether chemotherapy is successful is its schedule: even with the same total dose, the cadence at which it's taken can drastically change the results.

Figure 5.4 shows the tumor population's behaviour when following a particular chemotherapy schedule where there is a 3-day gap between each cycle. We see that the immune system can kill the remaining tumor cells after chemotherapy.

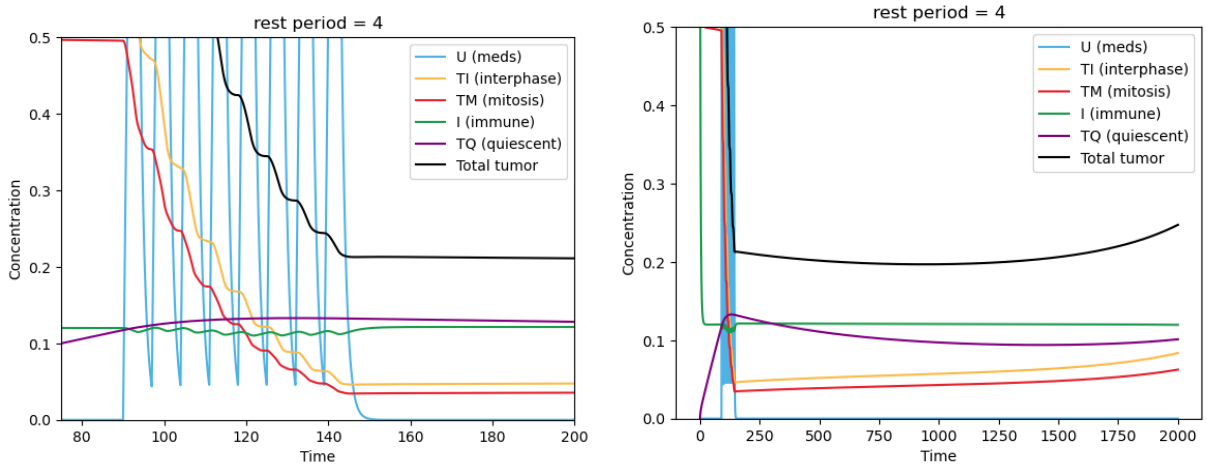


(a) Behaviour during chemotherapy

(b) Long-term behaviour

Figure 5.4: Simulation of chemotherapy with 3 days of medication followed by 3 days of rest

Figure 5.5 shows the behaviour when the gap is increased to 4 days. We see that the concentration of the drug reaches a much lower trough compared to the previous case, resulting in long-term behaviour where the cancer cells can overpower the immune cells.



(a) Behaviour during chemotherapy

(b) Long-term behaviour

Figure 5.5: Simulation of chemotherapy with 3 days of medication followed by 4 days of rest

This suggests that administering chemotherapy doses in close succession can enhance its efficacy and combat tumor faster. However, caution must be taken since chemotherapy also harms normal cells, and an overly aggressive may harm the patient instead.

Simulation details for this analysis:

- $\tau_q = 30$ was arbitrarily chosen.
- The chemotherapy regiment simulated was $8\times$(3 days of chemotherapy + 3 or 4 days of rest), where u reaches 0.5 (arbitrary choices).
- Initial conditions $(T_I, T_M, I, u, T_Q) = (2.5, 2.5, 0.9, 0, 0)$ was chosen arbitrarily.
- All other parameter values are identical to Table 2.1.

Chapter 6

Conclusion

This work investigated tumor growth using a mathematical model, incorporating the dynamics of QCCs through an extension. We employed a DDE model to track key cell populations and analyze their response to cytotoxic drugs. The stability analysis focused on the tumor-free steady state, a critical indicator of treatment effectiveness. We further explored the basin of attraction to identify treatment regimens that effectively suppress tumor growth.

The inclusion of QCCs significantly impacts the model's behaviour, highlighting the importance of considering this subpopulation in treatment strategies. The model suggests that therapies targeting both proliferating and quiescent cancer cells are necessary for long-term tumor control.

This framework offers potential for further development:

1. ***Incorporating Additional Complexities:*** Future work could explore additional biological complexities such as spatial tumor heterogeneity or the immune system's role in controlling QCCs.

2. ***Patient-Specific Modeling:*** The model could be parameterized using patient-specific data to guide personalized treatment plans.

Overall, this study demonstrates the value of mathematical modeling in understanding tumor growth dynamics. By incorporating biological complexities like QCCs, the model provides a more comprehensive framework for evaluating treatment strategies and paving the way for potential advancements in cancer therapy.

References

[1] Minaya Villasana and Gabriela Ochoa. Heuristic design of cancer chemotherapies. *IEEE Transactions on Evolutionary Computation*, 8(6):513–521, December 2004. ISSN 1089-778X. doi: 10.1109/TEVC.2004.834154. URL https://ieeexplore.ieee.org/document/1369244/.

[2] Joy Pader, Yibing Ruan, Abbey E. Poirier, Keiko Asakawa, Chaohui Lu, Saima Memon, Anthony Miller, Stephen Walter, Paul J. Villeneuve, Will D. King, Karena D. Volesky, Leah Smith, Prithwish De, Christine M. Friedenreich, and Darren R. Brenner. Estimates of future cancer mortality attributable to modifiable risk factors in Canada. 112(6):1069–1082, May 2021. ISSN 0008-4263, 1920-7476. doi: 10.17269/s41997-020-00455-7. URL https://link.springer.com/10.17269/s41997-020-00455-7.

[3] Bruce Alberts, Alexander Johnson, Julian Lewis, Martin Raff, Keith Roberts, and Peter Walter. *Molecular Biology of the Cell*. Garland Science, 4th edition, January 2002. ISBN 978-0-8153-3218-3 978-0-8153-4072-0. doi: 10.1201/9780203833445.

[4] Stavros N. Busenberg and Kenneth L. Cooke, editors. *Differential Equations and Applications in Ecology, Epidemics, and Population Problems*. Academic Press, New York, NY, January 1981. ISBN 978-0-12-148360-9. doi: 10.1016/B978-0-12-148360-9.X5001-X.

[5] Silviu-Iulian Niculescu, Islam Boussaada, Xu-Guang Li, Guilherme Mazanti, and César-Fernando Méndez-Barrios. Stability, Delays and Multiple Characteristic Roots in Dynamical Systems: A Guided Tour. In *Proceedings of the 16th IFAC Workshop on Time Delay Systems*, volume 54 of *TDS 2021*, pages 222–239. IFAC-PapersOnLine, October 2021. doi: 10.1016/j.ifacol.2021.11.143. URL https://linkinghub.elsevier.com/retrieve/pii/S2405896321021728.

[6] Angela Zaccagnino, Antonella Managò, Luigi Leanza, Artur Gontarewitz, Bernhard Linder, Michele Azzolini, Lucia Biasutto, Mario Zoratti, Roberta Peruzzo, Karen Legler, Anna Trauzold, Holger Kalthoff, and Ildiko Szabo. Tumor-reducing effect of the clinically used drug clofazimine in a SCID mouse model of pancreatic ductal adenocarcinoma. *Oncotarget*, 8(24):38276–38293, June 2017. ISSN 1949-2553. doi: 10.18632/oncotarget.11299. URL https://oncotarget.com/lookup/doi/10.18632/oncotarget.11299.

[7] Joseph D. Butner, Prashant Dogra, Caroline Chung, Renata Pasqualini, Wadih Arap, John Lowengrub, Vittorio Cristini, and Zhihui Wang. Mathematical modeling of cancer immunotherapy for personalized clinical translation. *Nature Computational Science*, 2(12):785–796, December 2022. ISSN 2662-8457. doi: 10.1038/s43588-022-00377-z. URL https://nature.com/articles/s43588-022-00377-z.

[8] Minaya Villasana and Ami Radunskaya. A delay differential equation model for tumor growth. *Journal of Mathematical Biology*, 47(3):270–294, May 2003. ISSN 0303-6812, 1432-1416. doi: 10.1007/s00285-003-0211-0. URL http://link.springer.com/10.1007/s00285-003-0211-0.

[9] Kenneth L. Cooke and Pauline van den Driessche. On Zeroes of Some Transcendental Equations. *Funkcialaj Ekvacioj*, 29(1):77–90, 1986. doi: 10.24546/0100499497. URL http://fe.math.kobe-u.ac.jp/FE/FullPapers/vol29/fe29-1-6.pdf.

[10] John E. Davis Jr, Jason Kirk, Yibing Ji, and Dean G. Tang. Tumor Dormancy and Slow-Cycling Cancer Cells. In Johng S. Rhim, Anatoly Dritschilo, and Richard Kremer, editors, *Human Cell Transformation*, volume 1164 of *Advances in Experimental Medicine and Biology*, pages 199–206. Springer International Publishing, Cham, Switzerland, October 2019. ISBN 978-3-030-22253-6 978-3-030-22254-3. doi: 10.1007/978-3-030-22254-3_15. URL http://link.springer.com/10.1007/978-3-030-22254-3_15.

APPENDICES

Appendix A

Python Code

```
# pre-existing model parameters
tau = 0.92
a1 = 0.98
a4 = 0.8
d2 = 0.11

d3 = 0.4
d1 = 0.29
c1 = 0.9 #9e-1
c3 = 0.9 #9e-1

c2 = 85e-3
c4 = 85e-3
rho = 0.1
alpha = 0.2

k=0.036
n = 3
lambda1 = 126.12
lambda2 = 0.85

k1 = 0.47
k2 = 0.57
k3 = 0.49
k4 = 0.061

```

```python
r1 = 0.73
r2 = 0.27

# custom parameters
aq = 0.00125 # production term for TM->TQ
dq = 0.00125 # exit term for TQ->TI
tau_q = 30

u_target = 0.5
u_active = 3
u_rest = 4
u_cycles = 8
u_start = 90

u_cycle_duration = u_active + u_rest # do not change

def delay_rhs(x, t):
    global dq
    TI, TM, I, u, TQ = x(t)
    TI_past, _, _, _, _ = x(t - tau)
    _, _, _, _, TQ_past= x(t - tau_q)


    dx = [0,0,0,0,0]
    dx[0] = 2*a4*TM - (c1*I + d2)*TI - a1*(TI_past) + dq*(TQ_past)# TI
    dx[1] = a1*(TI_past) - d3*TM - a4*TM - c3*TM*I - k1*(1 - np.exp(-k2*u))*TM - aq*TM # TM

    if (alpha * 1000 < TI + TM):
        dx[2] = k + rho * I - c2*I*TI - c4*TM*I - d1*I - k3*(1 - np.exp(-k4 * u))*I# I
    else:
        dx[2] = k + (rho * I * (TI + TM)**n)/(alpha + (TI + TM)**n) - c2*I*TI - c4*TM*I - d1*I - k3*(1 - np.exp(-k4 * u))*I# I

    dx[3] = - lambda2 * u # u
    if ( t > u_start and t < u_start + u_cycle_duration * u_cycles and (t - u_start) % u_cycle_duration < u_active):
        dx[3] = u_target

```

```
    dx[4] = 2*aq*TM - dq*(TQ_past) # TQ

    for i in range(4):
        if x(t)[i] + dx[i] >= 100 :
            dx[i] = 0

    return dx

# Initial history
def initial_history_func(t):
    # TI TM I u TQ
    return [2.5, 2.5, 0.9, 0, 0]

# Time points we want
t = np.linspace(0, 2000, 40000)

# Perform the integration
x = ddeint(delay_rhs, initial_history_func, t)

plt.figure()
plt.plot(t, x[:,3], label="U␣(meds)", color='#4eb3e4')
plt.plot(t, x[:,0], label="TI␣(interphase)", color='#fabb4b')
plt.plot(t, x[:,1], label="TM␣(mitosis)", color='#e6222e')
plt.plot(t, x[:,2], label="I␣(immune)", color='#1d9a4d')
plt.plot(t, x[:,4], label="TQ␣(quiescent)", color='purple')
plt.plot(t, x[:,0]+ x[:,1]+ x[:,4], label="Total␣tumor", color='black')

plt.xlabel("Time")
plt.ylabel("Concentration")
plt.title(f"rest␣period␣=␣{u_rest}")
plt.legend()
plt.ylim(0,0.5)
plt.show()
```

Glossary

basin of attraction The region of the phase space, over which iterations are defined, such that any point (any initial condition) in that region will asymptotically be iterated into the attractor. 13

chemotherapy Treatment that uses drugs to stop the growth of cancer cells, either by killing the cells or by stopping them from dividing. 5

cytotoxic Toxic to cells. 1

deterministic system A system in which no randomness is involved in the development of future states. 8

growth arrest A stopping point in the cell cycle, where it is no longer involved in the processes surrounding duplication and division. 5

half-life The length of time required for the concentration of a particular substance (typically a drug) to decrease to half of its starting dose in the body. 5

Hopf bifurcation The local birth or death of a periodic solution (self-excited oscillation) from equilibrium as a parameter crosses a critical value. 8

Michaelis-Menten term Mainly used to characterize the enzymatic rate at different substrate concentrations, but it is also widely applied to characterize the elimination of chemical (the first-order kinetics) compounds from the body. 6

relapse Slip back into a previous state of illness. 1

spatial heterogeneity The existence of differences in characteristics such as genetic information and cell morphology. 8

stochastic Involving a random variable. 8